PULSE / SIMPLE IR REMOTE

DIRECT GPIO DRIVE / ONE EMITTER / MINIMAL COMPONENTS

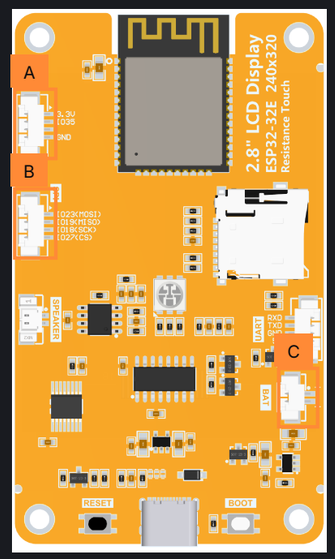
E32R28T

ASSEMBLY SHEET

01 / E32R28T CONNECTIONS

REAR VIEW

Antenna up / USB-C down



A / P4

1	3V3
2	IO35 / RX
3	NC - leave open
4	GND

B / P3

1	IO23 / MOSI
2	IO19 / MISO
3	IO18 / SCK
4	IO27 / TX

C / BATTERY

Top: BAT +

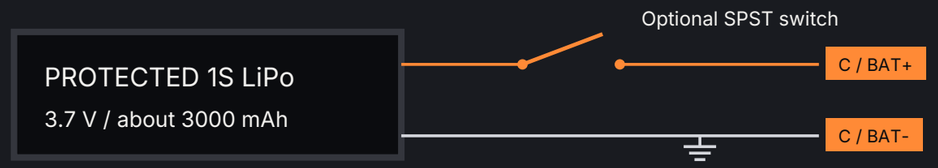
Bottom: BAT -

Pin order is top to bottom in this rear view, NOT the mating plug view. Verify silkscreen and polarity.

Use matching 1.25 mm connector pigtails. Ordinary 2.54 mm Dupont plugs do not fit these sockets.

GPIO34 is already wired to the onboard 100k / 100k battery divider. Do not connect BAT+ directly to it.

02 / BATTERY POWER - ONBOARD USB-C CHARGING

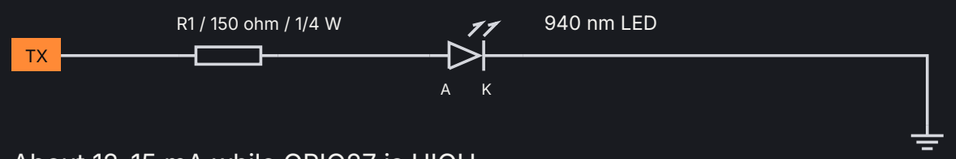


USB-C

Program + recharge onboard

Opening the battery switch stops battery charging. It does not turn the board off while USB is connected.

03 / TRANSMIT - ONE LED, NO MOSFET

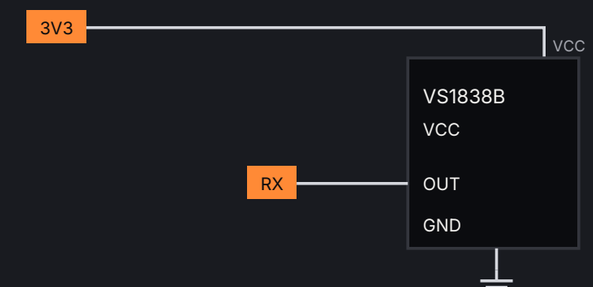


About 12-15 mA while GPIO27 is HIGH.

Use a standard 20 mA-rated emitter, or a TSAL6200. One LED concentrates the available GPIO drive in a single emitter; it does not increase the ESP32 current limit.

No battery feed to this LED. Never fit the full build's 33 ohm resistor here.

04 / RECEIVE - 3.3 V LOGIC



Logical pins, NOT physical leg order.
Check receiver markings / datasheet.

05 / FIRMWARE PROFILE

SIMPLE / one directly driven LED / nominal 50% carrier / no battery-only transmit lockout.

Flash the simple profile. Both builds support touch, IR learning, Favorites and eight saved-code slots.

06 / PARTS CHECKLIST

1 each	E32R28T; USB-C data cable; protected 1S LiPo + matching plug
1 + 1	940 nm, 20 mA-rated emitter + VS1838B/HX1838B receiver
1	150 ohm resistor, 1/4 W
none	No MOSFET, gate network or bulk capacitor required
optional	100 nF across receiver VCC/GND if learning is noisy
optional	SPST battery switch with adequate DC rating
as needed	1.25 mm pigtails, perfboard, insulated wire, enclosure and mounts

07 / ASSEMBLY + OPERATING NOTES

LED anode A is the long lead; cathode K is the flat-side lead. Connect GPIO27 -> 150 ohm -> A, then K -> GND. Always use the series resistor.

Receiver output must be 3.3 V logic. Keep its wires short; a 100 nF bypass capacitor is the first improvement if capture is noisy.

GPIO27 is separate from display and resistive-touch buses on this E32R28T. GPIO35 is input-only. Leave unused connector pins open.

Favorites and up to eight saved codes use onboard flash; no SD card is needed. Learning pauses during transmission. The ~38 kHz receiver captures IR timing, not carrier frequency or RF/Bluetooth signals.

Battery connector polarity and receiver lead order must be verified. Power off and unplug USB before wiring. The first touch after backlight sleep wakes without transmitting.

COMMON SOFTWARE / Touch UI, Favorites, eight saved codes; RX paused during TX; flash storage, no SD. ~38 kHz IR receiver, not RF/Bluetooth.

BOARD / PINOUT SCHEMATIC LED DATASHEET MOSFET DATASHEET IR SOFTWARE

POWER OFF BEFORE WIRING / VERIFY ON HARDWARE